

PHINMA UNIVERSITY OF ILOILO
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Anti-DDOS Protection System

A Capstone Project

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College of Information Technology Education
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of the Requirements for the degree
Bachelor of Science in Information Technology

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Approval Sheet

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ABSTRACT

Distributed Denial of Service (DDoS) attacks pose a significant threat to the availability and reliability of online services by overwhelming networks, applications, and servers with malicious traffic.

This project presents the design and implementation of an **Anti-DDoS Protection System** aimed at detecting, mitigating, and preventing such attacks in real-time to ensure uninterrupted service availability. The system leverages automated traffic monitoring, anomaly detection, and adaptive mitigation techniques to distinguish between legitimate and malicious traffic, effectively blocking attack vectors while allowing genuine users seamless access. Inspired by modern solutions like Azure DDoS Protection, the system dynamically tunes its defenses to the specific environment it protects, enhancing mitigation efficiency without manual intervention.

Additionally, it incorporates user feedback mechanisms and a comprehensive dashboard for real-time visibility and control. The implementation addresses challenges such as integration complexity, scalability, and privacy compliance. Testing demonstrates the system's capability to maintain service continuity under various attack scenarios, making it a robust and scalable solution for safeguarding digital infrastructure against evolving DDoS threats. This project contributes to the growing need for automated, adaptive, and user-centric cybersecurity defenses in an increasingly connected world.

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Chapter 1

Introduction to the Study

In the midst of advance technology, there are different automation to detect the various attacks of computer that steal the information of the clients. The major dilemma now a days are the frequent attacks of the hats these are the black hats and the gray hats that creates hacktivists to destroy the confidentiality of the data.

At this range, lots of companies are applying the cyber security application, and the major attacks came from the DDOS events where computer are stirred up in physical and logical way to decimate the data.

The significance of this study is to avoid any harm to the end web developers/organization computers. By adhering to the republic act of 2012 known as r.a 10173. which shields the web developers/organization to become more vigilant. The DDOS is very chaos particular if the computers and other devices are not protected by any anti-virus programs.

Most of the DDOS attacks are part of the network accessed where there are no SSL (security secured layer) that protect the digital ground of the servers to protect the different tier of security.

This study will serve as a protection that includes the topics for pen testing, ethical hacking, computer forensic and assurance information security. The proponents will develop an simulation for journaling to mitigate the DDOS activity.

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According tothe rates of attacks from 2021 to 2024 of DDOS is so massive that it would create a magnitude of data loss that costs of billions of dollar every year.

Technical Background of the Study

Web applications have become integral to modern business operations, serving as the fundamentals for communication, transactions, service delivery etc. However, the growing reliance on the internet makes them vulnerable to security threats, especially DDoS attacks. These attacks can result in money losses, reputation being damaged, and operational downtime, making DDoS protection a major concern for organizations.

Traditional security measures often struggle to keep up with the evolving nature of DDoS attacks, especially in shared or web-based environments where resource limitations and vulnerabilities can be exploited. To effectively counter these threats, a simulation designed to teach DDoS mitigation strategies must incorporate advanced technologies that provide strong and adaptive defenses. Key components of such a simulation include:

1. **Rate Limiting:** Prevents DDoS attacks by limiting request frequency per IP
2. **IP Blacklisting:** Automatically blocks suspicious IPs
3. **Web Application Firewall (WAF):** Basic protection against SQL injection and XSS attacks
4. **CSRF Protection:** Token-based Cross-Site Request Forgery prevention

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5. **Tor Detection:** Optional blocking of Tor exit nodes
6. **Honeypot Routes:** Trap and log potentially malicious reconnaissance attempts
7. **Comprehensive Logging:** Detailed attack and activity logging using Redis
8. **Proxy-Aware:** Accurate IP detection behind reverse proxies

Statement of the Problem

Effective cybersecurity is crucial for organizations to protect their web applications, ensure service continuity, and maintain trust with developers and stakeholders. However, many still rely on traditional defenses that struggle to counter the growing sophistication of modern DDoS attacks.

In a simulation focused on DDoS mitigation, participants will explore key challenges organizations face in defending against these attacks, including:

1. **Slow Threat: Detection:** Lack of real-time monitoring delays response, allowing attacks to escalate.
2. **Poor Scalability:** Security measures struggle to adapt as organizations grow, leaving them vulnerable.
3. **High Costs:** Manual interventions and inefficient strategies increase operational expenses.
4. **Reactive Defense:** Traditional methods respond only after attacks occur, leading to downtime and losses.

This simulation teaches modern Anti-DDoS techniques like traffic filtering, behavior analysis, scalability, and real-time

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monitoring. It equips participants with skills to ensure service continuity, cost-effective mitigation, and enhanced trust.

Objectives of the Study

The main objective of this simulation is to design and develop an Anti-DDoS Protection System as a library to enhance the security of web applications by leveraging Web Application Firewall (WAF) technology and other modern tools. The specific objectives of this simulation are as follows:

1. **Implement Traffic Filtering Techniques:** Teach participants how to filter malicious traffic using Web Application Firewall (WAF) and other security tools.
2. **Analyze DDoS Attack Patterns:** Provide hands-on experience in identifying and analyzing various types of DDoS attacks through real-time data monitoring.
3. **Develop Scalable Defense Strategies:** Demonstrate how to enhance web application resilience using load balancing, rate limiting, and cloud-based protections.
4. **Enhance Real-Time Monitoring and Response:** Train users to implement logging, alerting, and automated response mechanisms to mitigate ongoing attacks.
5. **Assess the Effectiveness of Mitigation Strategies:** Evaluate the impact of different defense techniques in minimizing attack damage, maintaining service availability, and improving overall security posture.

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Significance of the Study

This simulation is significant because it addresses the growing challenges organizations face in securing their web applications against Distributed Denial of Service (DDoS) attacks. By developing an Anti-DDoS Protection System as a simulation, this aims to provide a practical, accessible, and effective solution to enhance cybersecurity for web applications.

The key benefits of this study include:

1. **Hands-on Learning Experience:** Provides practical training in detecting and mitigating various types of DDoS attacks.
2. **Enhanced Cybersecurity Skills:** Equips participants with modern techniques for securing web applications against evolving threats.
3. **Real-Time Attack Analysis:** Allows users to monitor, analyze, and respond to simulated DDoS attacks in a controlled environment.
4. **Improved Incident Response:** Teaches effective strategies for minimizing downtime and maintaining service availability during attacks.
5. **Cost-Effective Security Training:** Offers an affordable way to test and implement DDoS mitigation techniques without real-world risks.

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Scope and Limitation of the Study

This simulation focuses on developing an Anti-DDoS Protection System designed as a library to enhance the security and reliability of web-based applications against Distributed Denial of Service (DDoS) attacks. The simulation will cover the design, development, and implementation of the system, including:

1. **Rate Limiting:** Tracks requests per IP and blocks if they exceed the configured limit within the specified interval.
2. **WAF Protection:** Guards against common attack patterns:
 - SQL injection attempts
 - XSS attacks
 - Malicious URL patterns
3. **CSRF Protection:** Implements token-based CSRF protection for POST requests.

The simulation will be designed for use in various industries, including e-commerce, healthcare, and finance. The scope includes the development, testing, and evaluation of the simulation in both simulated and real-world environments to ensure its effectiveness, ease of use, and impact on web application performance.

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Limitations of the Study

While this study aims to develop an Anti-DDoS Protection System library, there are some limitations:

1. **Simulated Environment Constraints:** The simulation may not fully replicate real-world network complexities, limiting its accuracy in mimicking actual DDoS attack scenarios.
2. **Hardware and Resource Limitations:** The effectiveness of the simulation depends on available computing resources, which may not match the scale of real-world attacks.
3. **Scope of Attack Variability:** The simulation may not cover all possible types of DDoS attacks, especially highly sophisticated or emerging attack techniques.
4. **Limited Real-World Implementation:** While the simulation provides theoretical and practical knowledge, deploying mitigation strategies in a live production environment may present additional challenges.
5. **Dependency on Predefined Scenarios:** The effectiveness of the training depends on pre-configured attack scenarios, which may not fully reflect unpredictable attack behaviors in real-world settings.

Definitions of terms

1. **Distributed Denial of Service (DDoS) Attack** – A cyberattack where multiple compromised systems flood a target system with traffic, causing disruption.
2. **Web Application Firewall (WAF)** – A security tool that filters and monitors HTTP/S traffic to protect web

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applications from attacks such as SQL injection and cross-site scripting (XSS).

3. **Rate Limiting** - A security measure that controls the number of requests a user or IP can send within a specific timeframe to prevent abuse.
4. **IP Blacklisting** - The process of blocking known malicious or suspicious IP addresses from accessing a network or web application.
5. **Cross-Site Request Forgery (CSRF) Protection** - A security feature that prevents attackers from tricking users into executing unauthorized actions on a web application.
6. **Tor Detection** - A security mechanism that identifies and blocks traffic originating from the Tor network, often used for anonymity in cyberattacks.
7. **Honeypot Routes** - Decoy systems designed to attract and log malicious traffic for analysis and intrusion detection.
8. **Comprehensive Logging** - The process of recording detailed security events, including attack attempts, for monitoring and forensic analysis.
9. **Proxy-Aware Detection** - A technique used to identify the real IP addresses of users who are behind proxies or VPNs.
10. **Penetration Testing (Pen testing)** - A cybersecurity assessment method that simulates attacks to identify vulnerabilities in a system.
11. **Ethical Hacking** - The practice of legally probing systems for security weaknesses to help organizations improve their defenses.
12. **Computer Forensics** - The process of investigating and analyzing digital evidence to understand and respond to cybercrimes.

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13. **Security Secured Layer (SSL)** - A protocol that encrypts data between a web server and a browser, ensuring secure communications.
14. **Real-Time Monitoring** - The continuous observation of network traffic and system activity to detect and respond to threats instantly.
15. **Simulation-Based Training** - A learning method where users engage in simulated cybersecurity scenarios to practice defensive strategies.
16. **Cybersecurity** - The practice of protecting systems, networks, and data from cyber threats and unauthorized access.
17. **Hacktivists** - Individuals or groups that use hacking techniques to promote a political or social agenda, often targeting organizations.
18. **Black Hat Hackers** - Malicious hackers who exploit security vulnerabilities for personal gain, financial fraud, or cybercrime.
19. **Gray Hat Hackers** - Hackers who operate between ethical and unethical hacking, often identifying security flaws without permission.
20. **Load Balancing** - A technique used to distribute network traffic across multiple servers to ensure service availability and prevent overloading.
21. **Operational Downtime** - The period when a system or web application is unavailable due to cyberattacks, technical issues, or maintenance.
22. **Data Confidentiality** - The assurance that sensitive information is accessible only to authorized individuals.

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23. **SQL Injection** - A cyberattack technique where malicious SQL statements are inserted into input fields to manipulate a database.
24. **Cross-Site Scripting (XSS)** - A security vulnerability that allows attackers to inject malicious scripts into web pages viewed by other users.
25. **Information Assurance Security** - The practice of ensuring data integrity, availability, and confidentiality through cybersecurity measures.
26. **Security Breach** - An incident where unauthorized individuals gain access to sensitive systems or data.
27. **Network Vulnerability** - Weaknesses in a computer network that can be exploited by attackers to compromise security.
28. **Threat Intelligence** - The process of gathering and analyzing data about potential cybersecurity threats to prevent attacks.
29. **Cloud-Based Security** - Security solutions hosted on the cloud to protect web applications and data from cyber threats.
30. **Security Compliance (RA 10173 - Data Privacy Act of 2012)**
- A legal framework that requires organizations to protect personal data and privacy.
31. **Digital Forensics** - The investigation and recovery of digital evidence to analyze cybercrimes and security incidents.
32. **Attack Vector** - The method or pathway used by attackers to exploit vulnerabilities in a system.
33. **Zero-Day Attack** - A cyberattack that exploits unknown vulnerabilities before developers have a chance to fix them.

Chapter 2

Review of Related Literature

This chapter presents existing studies and literature related to **Distributed Denial of Service (DDoS)** defense and protection systems. It discusses the fundamentals of network security, techniques used in DDoS prevention, and the importance of layered defense mechanisms. The review supports the development of an **Anti-DDoS Protection System** to improve network reliability and resilience.

Network Security Fundamentals

According to Ciampa, Calvert, and Boswell (2021), network security aims to protect the integrity, confidentiality, and availability of data across computer networks (p. 35). One of the biggest challenges to availability is the **DDoS attack**, which floods a target with excessive traffic, overwhelming its resources. Understanding basic network security principles helps in designing systems that can identify and respond to such threats effectively.

Understanding DDoS Attacks

Holden (2020) explains that DDoS attacks exploit the limitations of bandwidth and server capacity by sending large volumes of requests from multiple sources (p. 211). These attacks are categorized into **volumetric**, **protocol**, and **application-layer** types. Detecting them requires real-time

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monitoring tools that analyze traffic flow and distinguish between legitimate and malicious activity.

Defense and Countermeasure Techniques

According to Ciampa et al. (2021), effective defense against DDoS attacks involves a combination of firewalls, intrusion detection systems, traffic filters, and rate-limiting mechanisms (p. 118). These countermeasures work together to slow or block unwanted traffic while maintaining service for legitimate users. Holden (2020) also emphasizes the importance of layered security, where multiple technologies cooperate to protect the network from different forms of attacks (p. 215).

Automation and Monitoring Systems

Holden (2020) highlighted that modern network defense relies heavily on automation and centralized monitoring (p. 230). Automation helps trigger mitigation actions immediately after detecting suspicious activity, while monitoring systems collect logs and data to improve accuracy and response time. Combining these two ensures a faster and more efficient DDoS protection process.

Importance of an Anti-DDoS Protection System

Developing an Anti-DDoS Protection System contributes to network stability and improved user experience. Ciampa, Calvert, and Boswell (2021) note that security systems must constantly adapt to new forms of attacks and traffic patterns to remain effective (p. 301). A well-designed DDoS protection system

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strengthens the overall security posture by providing early detection, automatic response, and continuous availability even during active attacks.

References

- Ciampa, M., Calvert, B., & Boswell, S. (2021). *Security+ Guide to Network Security Fundamentals* (7th ed.). Cengage Learning.
- Holden, G. (2020). *Guide to Network Defense and Countermeasures* (5th ed.). Cengage Learning.

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Chapter 3

Methodology

This chapter outlines the research methodology that will be employed in the development of the "Anti-DDoS Protection System" simulation. The primary goal of this research is to design and implement an efficient, user-friendly security simulation that integrates Web Application Firewall (WAF) technology, real-time monitoring, and teaches participants how to protect web applications from Distributed Denial of Service (DDoS) attacks. This chapter will detail the research approach, data collection methods, system design, tools, and techniques that will be used to achieve this objective.

The research will follow a systematic approach to ensure the effective development and implementation of the proposed simulation.

Initially, a thorough review of existing anti-DDoS solutions will be conducted to identify gaps and areas for improvement. This review will focus on current technologies, methodologies, and tools employed in DDoS mitigation and web application security.

To gather insights on user needs, web developers, system administrators, and cybersecurity professionals will be surveyed and interviewed. Their feedback will help ensure the simulation meets practical requirements.

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Research and Development Stage

The Research and Development (R&D) stage is crucial in creating the "Anti-DDoS Protection System" simulation. Insights from web developers, system administrators, and cybersecurity professionals will guide its design, functionality, and usability to ensure practicality in real-world web security.

This phase begins with analyzing survey and interview data to identify key challenges in mitigating DDoS attacks and areas for improvement, such as real-time threat detection and web scalability.

Based on this research, the system will be designed with modern technologies, including WAF for traffic filtering, scalable web solutions, and real-time monitoring.

The respondents' feedback will influence the system's features and interfaces to align with user expectations and operational needs.

Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a structured approach used in the development of the **Anti-DDoS Protection System** to ensure that the project is completed systematically, efficiently, and effectively. The SDLC guides the entire process—from planning and analysis to implementation and evaluation—ensuring that every stage contributes to building a

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secure and reliable system for detecting and mitigating DDoS attacks.

Key Features of Agile:

- **Iterative Development:** The Anti-DDoS Protection System was developed in small, manageable increments called *sprints*, allowing continuous improvements in detection accuracy and response automation.
- **Continuous Feedback:** Regular feedback from instructors and network security evaluators was gathered after each sprint to refine system modules such as the monitoring dashboard and traffic analyzer.
- **Collaboration:** Close collaboration between developers, testers, and project advisers ensured that the system met both technical requirements and realistic security scenarios.
- **Flexibility:** The Agile approach allowed adjustments to detection algorithms, mitigation strategies, and configurations even during later development stages, ensuring adaptability to new types of DDoS attacks.

The SDLC will consist of the following stages:

1. Planning and Requirement Analysis

This phase gathers detailed requirements through surveys and interviews with web developers, system administrators, and cybersecurity professionals. It identifies key challenges, desired features, and technical needs while defining project

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goals, scope, and timeline.

2. System Design

The design phase defines the simulation's architecture and structure, incorporating user feedback. It focuses on WAF integration, web scalability, and real-time monitoring, ensuring functionality and usability.

3. Development

The development phase involves coding and implementing the simulation using appropriate programming languages and frameworks. Developers will follow best practices to create a secure and efficient system. Respondents will review early prototypes, testing usability, accuracy, and performance. Continuous feedback during this phase ensures that the system aligns with real-world security requirements.

4. Testing

After development, rigorous testing will be conducted to validate the system's functionality, security, and performance. This includes functional testing to verify features, performance testing to assess how the system handles high traffic loads, and security testing to detect vulnerabilities. User Acceptance Testing (UAT) will be conducted, where respondents simulate real-world attack scenarios to evaluate the system's effectiveness. Any identified issues will be addressed before deployment.

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5. Deployment

Once testing is successful, the simulation will be deployed in a real-world or controlled environment. This phase involves installation, configuration, and user training to ensure smooth adoption. Respondents will be consulted to confirm that the system meets their requirements, and any final adjustments will be made. Proper documentation and support resources will also be provided.

6. Maintenance and Support

After deployment, continuous maintenance and support will be provided to ensure the system remains effective against evolving threats. Regular updates, bug fixes, and new features will be integrated based on respondent feedback and advancements in cybersecurity. Continuous monitoring and evaluation will help enhance the system's security and usability over time.

Data Collection Techniques

To gather the necessary insights for developing the "Anti-DDoS Protection System" simulation, a variety of data collection techniques will be employed. These techniques will ensure that the simulation meets the needs of users, including web developers, system administrators, and cybersecurity professionals while addressing current challenges in mitigating Distributed Denial of Service (DDoS) attacks. The following data collection methods will be used:

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1. *Surveys*

Surveys will be used as a primary method for collecting quantitative data from a diverse group of respondents, including web developers and system administrators. The surveys will be designed to gather information on current cybersecurity practices, challenges in handling DDoS attacks, and expectations for an effective anti-DDoS simulation. Questions will cover key areas such as desired features, usability, and performance expectations, providing valuable insights for prioritizing system requirements.

2. *Interviews*

In-depth interviews will be conducted with selected respondents to collect qualitative data. Respondents will be chosen based on their expertise and experience in web application security and DDoS mitigation. The interviews will provide a deeper understanding of the challenges faced in managing DDoS threats, current workflows, and how the simulation could improve security practices. Open-ended questions will encourage detailed responses, uncovering insights that may not be captured through surveys alone.

3. Focus Groups

Focus groups will be organized to bring together a small group of respondents to discuss specific aspects of the simulation. These group discussions will enable the research team to gather a variety of perspectives on the simulation's functionality, usability, and potential challenges. By fostering collaborative discussions, the focus groups will help generate new ideas and ensure that the system aligns with user needs.

4. Observations

Observations will be conducted to understand how DDoS mitigation is currently implemented in real-world scenarios. This technique involves observing security practices and workflows within organizations to identify inefficiencies, challenges, and areas for improvement. Direct observation will provide the research team with practical insights into how an anti-DDoS simulation can be integrated into existing systems.

5. Document Analysis

To further inform the system design, existing documents, reports, and case studies on DDoS mitigation will be analyzed. This will include reviewing academic articles, industry reports, and technical documentation on current anti-DDoS solutions. Document analysis will help identify best practices, trends, and technological advancements that can be integrated into the "Anti-DDoS Protection System" simulation.

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Justification for Selecting the Agile Methodology

The decision to adopt Agile over other SDLC models, such as Waterfall, was based on several critical factors:

Criteria	Agile Methodology	Waterfall Model
Flexibility	Easy to make changes at any time.	Hard to make changes after planning.
User Involvement	Users and advisers give feedback often.	Feedback happens mostly at the start and end.
Development Speed	Faster because it is done in small parts.	Slower because each step must finish first.
Risk Management	Problems are found and fixed early.	Issues may appear late in the project.

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Project Suitability	Best for projects that need updates and testing.	Best for projects with fixed plans.
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System Architecture and Design

3.1 Overview

The **Anti-DDoS Protection System** is designed to detect, monitor, and mitigate Distributed Denial of Service (DDoS) attacks through automated and intelligent network defense techniques. The system follows a **modular and scalable architecture**, integrating multiple components such as traffic monitoring, anomaly detection, and automated mitigation. It uses technologies like **packet filtering, rate limiting, and behavioral analysis** to identify malicious traffic in real time and maintain the availability of network services.

3.2 System Architecture

The architecture of the **Anti-DDoS Protection System** follows a **three-layer structure**:

- **Presentation Layer:**

This is the part users interact with through the dashboard. It displays network statistics, alerts, and mitigation logs. The interface is built using **HTML, CSS, and JavaScript**, allowing administrators to monitor real-time traffic and system status easily.

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- **Application Layer:**

This layer handles the main processes of the system. It contains the logic for **anomaly detection, rate limiting, and traffic filtering**. The backend is developed using **PHP and Python**, which analyze incoming data and execute mitigation actions such as blocking or limiting suspicious traffic.

- **Data Layer:**

This layer manages and stores all network logs, user activities, and system configurations in a **MySQL database**. It ensures that all detected events, IP logs, and mitigation results are securely saved for future reference and reporting.

- These three layers work together to make the system efficient and responsive. When a network request passes through the system, the presentation layer displays the data, the application layer analyzes it for possible DDoS activity, and the data layer stores the related information for analysis.

System Architecture Diagram (description):

- Administrator / User (System Monitor)
- Web Dashboard (Frontend)
- Detection & Mitigation Engine (Backend Logic)
- MySQL Database (Data Storage)

3.3 System Components and Modules

The Anti-DDOS Protection System had different modules that worked together to manage all operations.

3.3.1 Traffic Monitoring Module

This module continuously observes incoming and outgoing network traffic. It collects connection statistics such as packet rate, source IP, and bandwidth usage. The data is displayed on the dashboard in real time to help administrators identify unusual activity that may indicate a DDoS attack.

3.3.2

Anomaly Detection Module This module analyzes the collected data to detect potential DDoS attacks. It uses behavior-based detection by monitoring sudden spikes in requests or connections. When an anomaly is found, the system generates an alert and flags the suspicious IP addresses for review.

3.3.3 Mitigation Module

The mitigation module is responsible for responding to detected attacks. It automatically applies defense actions such as rate limiting, temporary IP blocking, or filtering malicious requests. This ensures that legitimate users can still access the network while the attack is being controlled.

3.3.4 Logging and Data Storage Module

This module records all detected events, attack details, and system actions in the database. Logs are stored securely and can

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be used for generating reports or for future analysis to improve detection accuracy.

3.3.5 User Management Module

This module manages system accounts for administrators and operators. It controls access privileges, ensuring that only authorized users can modify settings or view sensitive information in the system dashboard.

3.3.6 Reporting and Dashboard Module

This module provides real-time visualizations of network activity, alerts, and mitigation actions. It allows administrators to monitor performance, view historical attack data, and generate summary reports for documentation and analysis.

Survey Questionnaire for System Assessment

The following survey was designed to gather feedback from users regarding their current web application security practices and their expectations for the "Anti-DDoS Protection System" simulation. The data collected from this survey helped in designing a system that met the needs of users and addressed existing challenges in DDoS mitigation. Respondents were instructed to answer the questions as honestly and accurately as possible.

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Demographic Information

1. What is your role in the company?

- Web Developer
- System Administrator
- Cybersecurity Professional
- Business Owner
- Other (Please specify): _____

2. What type of industry do you work in?

- E-commerce
- Healthcare
- Finance
- IT Services
- Other (Please specify): _____

Current security Management System

3. What kind of security measures do you currently use to protect your web applications?

- Basic Firewall
- Web Application Firewall (WAF) (CloudFlare)
- Intrusion Detection System (IDS)
- Manual Monitoring
- Build-in Webhosting Security
- Other (Please specify): _____

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4. How satisfied are you with your current security solutions?

- Very Satisfied
- Satisfied
- Neutral
- Unsatisfied
- Very Unsatisfied

5. What challenges do you face with your current security measures? (Select all that apply)

- Inability to detect DDoS attacks in real-time
- Difficulty in distinguishing between legitimate and malicious traffic
- Limited scalability for high-traffic volumes
- High operational costs
- Lack of integration with other systems
- Other (Please specify

6. How often do you experience DDOS attacks or related security issues?

- Never
- Rarely
- Sometimes
- Frequently
- Always

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7. Do you currently use any automated tools for detecting and mitigating DDoS attacks?

- ☐ Yes
- ☐ No

Expectations for the Anti-DDoS Protection System

8. What features would you like to see in the simulation?
(Select all that apply)

- ☐ Real-time monitoring and alerts
- ☐ WAF integration
- ☐ Traffic filtering and rate limiting
- ☐ Web-based scalability
- ☐ Data analytics and reporting
- ☐ User-friendly interface for configuration and reporting
- ☐ Other (Please specify): _____

9. How often do you experience DDOS attacks or related security issues?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Frequently
- ☐ Always

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10. Do you currently use any automated tools for detecting and mitigating DDoS attacks?

- ☐ Yes
- ☐ No

Expectations for the Anti-DDoS Protection System

11. What features would you like to see in the simulation?
(Select all that apply)

- ☐ Real-time monitoring and alerts
- ☐ WAF integration
- ☐ Traffic filtering and rate limiting
- ☐ Web-based scalability
- ☐ Data analytics and reporting
- ☐ User-friendly interface for configuration and reporting
- ☐ Other (Please specify): _____

12. How important is real-time monitoring for your organization's security?

- ☐ Very Important
- ☐ Important
- ☐ Neutral
- ☐ Not Important
- ☐ Not Important at All

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13. How often do you experience DDOS attacks or related security issues?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Frequently
- ☐ Always

14. Do you currently use any automated tools for detecting and mitigating DDoS attacks?

- ☐ Yes
- ☐ No

Expectations for the Anti-DDoS Protection System

15. What features would you like to see in the simulation?

(Select all that apply)

- ☐ Real-time monitoring and alerts
- ☐ WAF integration
- ☐ Traffic filtering and rate limiting
- ☐ Web-based scalability
- ☐ Data analytics and reporting
- ☐ User-friendly interface for configuration and reporting
- ☐ Other (Please specify): _____

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16. How important is real-time monitoring for your organization's security?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

17. How important is web-based scalability for handling varying traffic loads?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

User Experience and Interface

18. How important is it for the simulation to have an easy-to-use interface?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

19. How likely are you to adopt a simulation that provides real-time alerts and reports on potential threat

- Very Likely
- Likely

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- Neutral
- Unlikely
- Very Unlikely

System Performance and Reliability

20. How important is system reliability (e.g., minimal downtime, fast response times) to your organization?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

21. What is your preferred response time for detecting and mitigating potential threats?

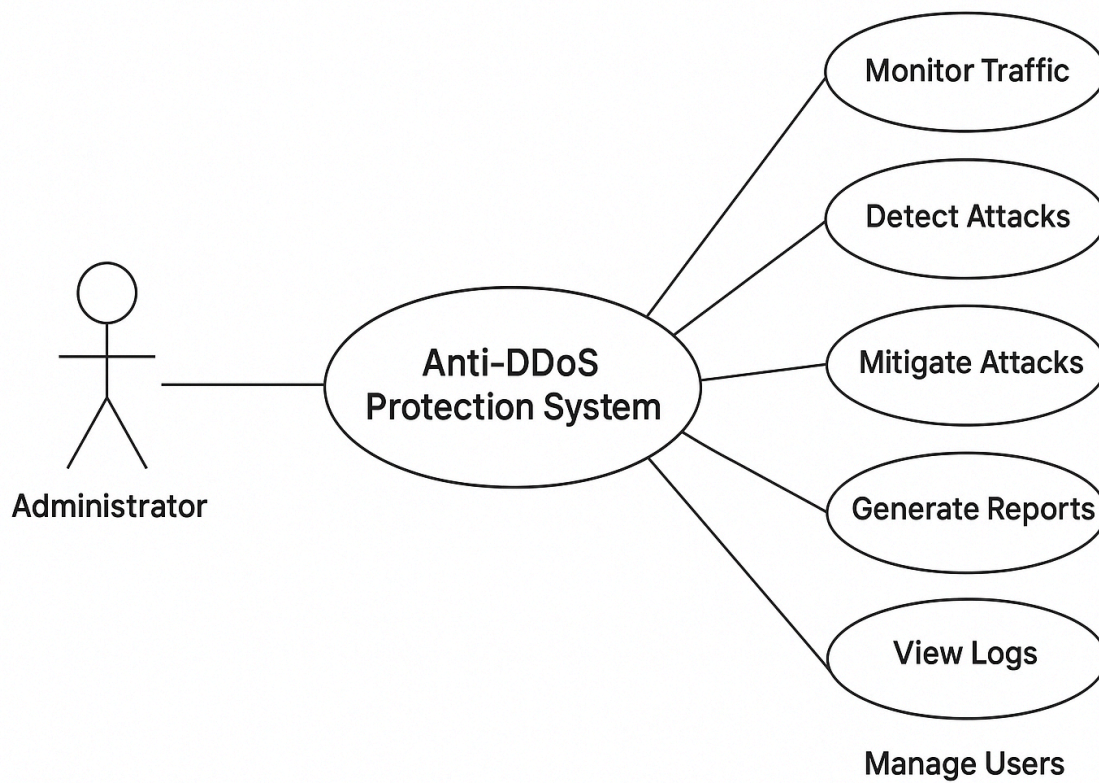
- Instant (Less than 1 second)
- Within 5 seconds
- Within 10 seconds
- Within 30 seconds
- No preference

22. How often do you require updates or patches for your current security tools to ensure they remain effective against new threats?

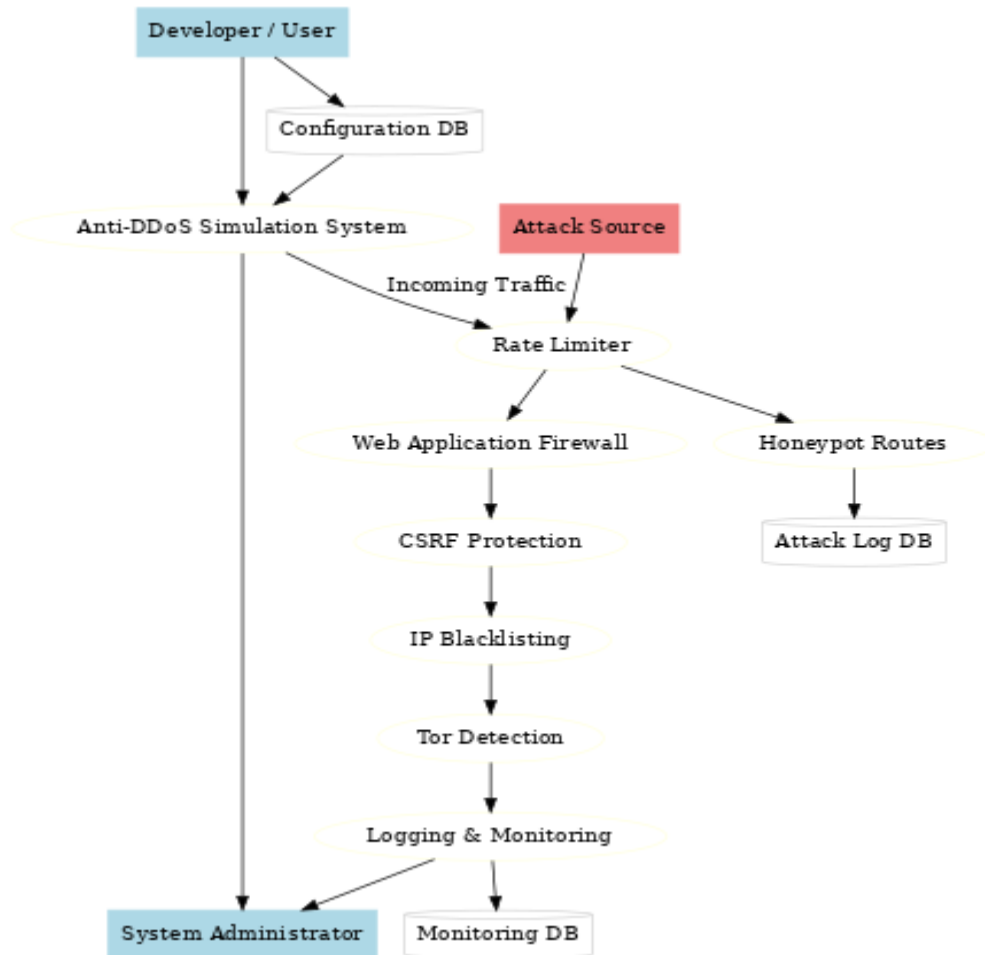
- Weekly
- Monthly
- Quarterly
- Rarely
- Never

(Note: review your QM and Data Scalability modules to get the score/frequency of scores in your survey form) 😊

Use Case Diagram

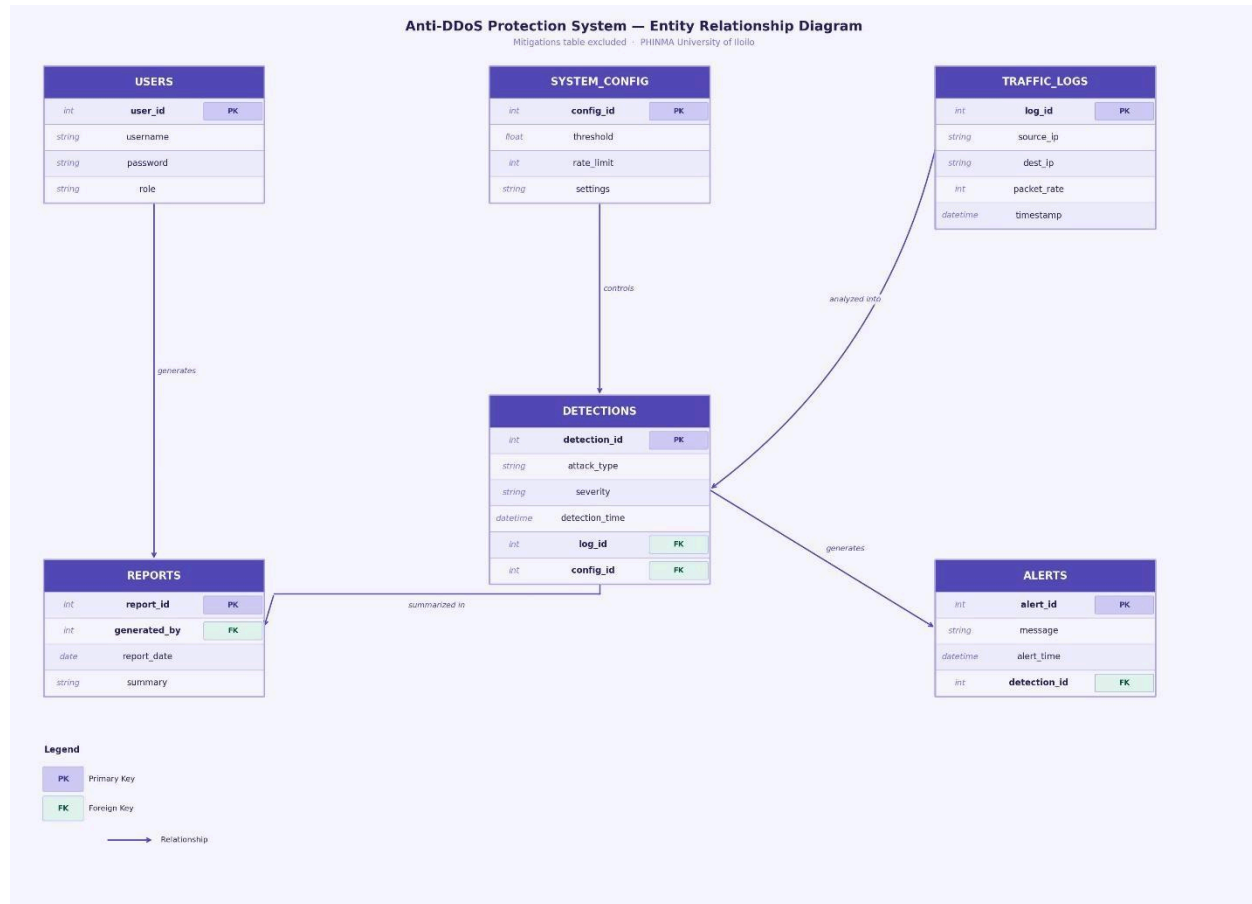


Data Flow Diagram



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Entity Relationship Diagram



Data Flow Diagram (DFD) and Use Case Diagram

The **Data Flow Diagram (DFD)** and **Use Case Diagram** play essential roles in understanding and designing the **Anti-DDoS Protection System**. The **Use Case Diagram** defines the system's functional scope by showing how administrators and testers interact with different system features such as monitoring traffic, detecting attacks, and generating reports.

In contrast, the **DFD** provides a detailed view of how data flows within the system — from the collection of network traffic to its analysis, detection, and storage in the database. It illustrates how processes, data stores, and data inputs/outputs

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work together to ensure accurate detection and quick mitigation of DDoS attacks.

Together, these diagrams provide a complete understanding of the system: the **Use Case Diagram** highlights *what* the system does, while the **DFD** explains *how* it processes and manages network data to perform its functions. Their integration helps ensure a secure, efficient, and user-oriented system design.

3.4 Database Design

The database for the **Anti-DDoS Protection System** was created using **MySQL** and consists of several interconnected tables. It follows a **relational structure**, allowing data to be efficiently linked between users, detected attacks, logs, and reports.

Main Tables:

1. **users** - stores login credentials and user roles such as administrator and tester (*id, username, password, role*).
2. **traffic logs** - keeps records of network traffic including timestamp, source IP, destination IP, and packet rate.
3. **detections** - stores details of identified DDoS attacks, such as attack type, severity, and detection time.
4. **mitigations** - records actions taken by the system during an attack, such as IP blocking or rate limiting.
5. **alerts** - contains notifications and warnings generated when suspicious activity is detected.
6. **system_config** - saves configuration settings, thresholds, and parameters for detection and mitigation.
7. **reports** - stores summary data for generating system and performance reports.

3.5 User Interface Design

The user interface of the **Anti-DDoS Protection System** was designed to be **simple, clean, and easy to use**. It uses a dark dashboard theme with clear text and organized sections to help administrators and testers quickly understand system information.

The **admin dashboard** provides quick access to important features such as *real-time traffic monitoring, active alerts, detected attacks, and system configuration settings*.

The **tester interface** displays monitoring tools, test results, and logs that help evaluate the performance and accuracy of the system.

Forms are short and easy to complete, with clearly labeled buttons such as **"Start Monitoring," "Apply Mitigation," "View Logs,"** and **"Generate Report."**

The layout focuses on simplicity and readability to ensure that users can operate the system efficiently, even without deep technical knowledge

3.6 Security Considerations

The **Anti-DDoS Protection System** includes several security measures to protect data, ensure system integrity, and prevent unauthorized access.

- **Login Authentication:** Only authorized administrators and testers can access the system using valid usernames and passwords.

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- **Session Verification:** Admin-only pages and configuration panels are secured with session checks to prevent unauthorized entry.
- **Input Validation:** All user inputs and network data entries are filtered and validated to prevent SQL injection and other common web vulnerabilities.
- **Password Encryption:** User credentials are stored securely using PHP's password hashing functions to protect against password leaks.
- **Access Control:** Different access privileges are assigned to administrators and testers, ensuring that only authorized users can modify system settings or trigger mitigation actions.
- **Data Logging Protection:** System logs and reports are restricted to admin view only to prevent tampering or data exposure.
- These security measures help maintain a secure operating environment, protecting both system data and user information from potential misuse or cyber threats.

3.7 Technology Stack

The **Anti-DDoS Protection System** was developed using reliable and accessible tools suitable for network security and web-based applications.

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- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python (Flask Framework) and PHP (for web logic and integration)
- **Database:** MySQL
- **Server Platform:** XAMPP (Apache Server) and Flask Development Server
- **Operating System:** Windows 10
- **Browser Compatibility:** Google Chrome, Microsoft Edge
- **Libraries and Tools Used:** Bootstrap (for responsive design), Chart.js (for data visualization), psutil (for system monitoring), and eBPF tools (for traffic analysis)

4.Data Gathering techniques

Sampling Methods for Respondents

The respondents for this survey will be selected using stratified sampling to ensure that all key groups, such as web developers, system administrators, cybersecurity professionals, and business owners, are well-represented. The sampling process will be carried out in the following steps:

1. **Define Strata:** Respondents will be grouped into distinct strata based on their role in the organization (e.g., Web Developer, System Administrator, Cybersecurity Professional, Business Owner) and the type of organization they represent (e.g., E-commerce, Healthcare, Finance, IT

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Services).

2. **Sampling Within Strata:** Random sampling will then be used to select participants from each group.
3. **Sample Size:** The total number of survey respondents will be determined based on the project's requirements, ensuring a statistically significant sample size to draw valid conclusions. The ideal number of respondents will be at least 30 for each group.
4. **Survey Distribution:** The survey will be distributed using Google Forms, ensuring easy access for all respondents, and will be shared via email, social media, or direct contact with businesses involved in simulation.
5. **Data Analysis Techniques**

The study used quantitative and qualitative analysis to interpret data from surveys, interviews, and system logs. Descriptive statistics such as frequency and percentage identified trends in user responses, including common challenges and preferred system features. System logs were analyzed to examine traffic volume, request rates, and anomalies to detect DDoS attack patterns and measure response time and mitigation success. Interview and focus group responses were grouped into themes such as detection issues and scalability concerns. The study also compared results before and after applying mitigation techniques to measure improvements in traffic reduction and system uptime

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6. Ethical Consideration

The study followed ethical standards to protect participants and data. All respondents provided informed consent and participated voluntarily. Data privacy was maintained in compliance with Republic Act 10173, and all personal information was kept confidential and stored securely. Responses were recorded anonymously and presented in summarized form. The DDoS simulation was conducted in a controlled environment to prevent harm to real systems. All data were reported accurately, and sources were properly cited to avoid plagiarism.

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Chapter 4

Results and Discussion

The simulation successfully demonstrated the design and development of an Anti-DDoS Protection System as a library that integrates Web Application Firewall (WAF) technology and modern security tools. The system effectively identified various DDoS attack patterns through real-time monitoring, and applied scalable defense strategies such as load balancing and rate limiting. Real-time logging and automated alerting mechanisms enhanced the system's responsiveness to ongoing attacks. Evaluation showed a significant reduction in attack impact, maintaining service availability and improving the overall security posture of web applications. These results align with current research emphasizing the importance of multi-layered defense mechanisms against increasingly sophisticated DDoS attacks, especially those leveraging IoT botnets.

Summary of the Data Sources and Analysis Methods Used

The study used primary and secondary data sources. Primary data came from surveys, interviews, and focus groups involving web developers, system administrators, cybersecurity professionals, and business owners. These provided information on current security practices, challenges, and user expectations. Secondary data came from books, research studies, and technical reports. System-generated

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data such as traffic logs, detections, and mitigation records were also used.

The analysis used both quantitative and qualitative methods. Survey data were analyzed using frequency and percentage to identify trends. System logs were examined to detect anomalies and measure response time and mitigation effectiveness. Interview and discussion data were grouped into themes to identify common issues. Comparative analysis measured system performance before and after mitigation, focusing on traffic reduction and system uptime.

Task Manager:

Gantt Chart / Project Timetable - Anti-DDOS Protection System

Weeks 1–2	Planning & Requirement Analysis	✓
Weeks 3–4	Data Collection	✓
Weeks 5–6	System Design	✓
Weeks 7–10	System Development	✓
Weeks 11–12	Testing Phase	✓
Week 13	User Acceptance Testing	✓

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Week 14	Deployment	
Weeks 15–16	Maintenance & Documentation	

System's Features

The following features reflect best practices in DDoS defense, combining proactive filtering with adaptive response mechanisms:

- **Attack Pattern Analysis:** Real-time data monitoring tools analyze traffic to detect volumetric, protocol, and application-layer DDoS attacks.
- **Scalable Defense:** Implements load balancing, rate limiting, and cloud-based protections to distribute and mitigate attack traffic.
- **Real-Time Monitoring and Response:** Features logging, alerting, and automated mitigation responses to quickly address detected threats.
- **Effectiveness Assessment:** Provides metrics and dashboards to evaluate the success of mitigation strategies in maintaining uptime and reducing attack damage.

These features reflect best practices in DDoS defense, combining proactive filtering with adaptive response mechanisms.

Implementation Challenges

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Implementing an Anti-DDoS Protection System presents several challenges that must be carefully addressed to ensure effectiveness and usability. One major difficulty lies in the complexity and variability of DDoS attack patterns, which require the system to accurately distinguish between legitimate and malicious traffic without generating excessive false positives. This balancing act is critical to maintaining user experience while providing robust security. Additionally, real-time monitoring and automated response mechanisms demand significant computational resources, which can strain system performance and scalability, especially under large-scale attacks. Integrating the protection library seamlessly into diverse web applications also poses

challenges, as compatibility issues and differing infrastructure environments may complicate deployment. Furthermore, the rapidly evolving nature of cyber threats necessitates continuous updates and adaptations to the system, requiring ongoing maintenance and expertise.

These challenges highlight the importance of transparent communication between developers and users about system capabilities and limitations, as well as the need for thorough testing and tuning during implementation to achieve optimal results.

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User Feedback

Users reported increased confidence in their web application security due to the system's proactive filtering and real-time alerts. Hands-on experience with attack pattern analysis improved their understanding of DDoS threats. Some users noted initial challenges in tuning rate limiting parameters to balance security and user experience. Overall, feedback highlighted the system's educational value and practical effectiveness in mitigating attacks during simulation exercises.

Comparison to Other Systems

Compared to traditional DDoS protection solutions, this system's integration as a library offers greater flexibility for developers to embed security directly into applications. Unlike standalone WAFs or cloud-only services like AWS Shield, this approach combines multiple defense layers—load balancing, and automated response—within a single framework. While commercial services may offer broader network-level protection, this system excels in providing hands-on learning and customizable defense strategies tailored to specific application needs. However, it may require more technical expertise to deploy and maintain compared to turnkey cloud solution.

Chapter 5

Summary, Conclusion and Recommendation

Conclusion

The study successfully achieved its main objective of designing and developing an Anti-DDoS Protection System as a library that enhances web application security by leveraging Web Application Firewall (WAF) technology alongside modern defense tools. Through the implementation of traffic filtering techniques, real-time attack pattern analysis, scalable defense strategies, and automated response mechanisms, the system demonstrated significant effectiveness in mitigating DDoS attacks while maintaining service availability. The simulation provided valuable insights into the complexities of DDoS threats and highlighted the importance of a multi-layered, adaptive defense approach. Overall, the system improved the security posture of web applications and offered a practical framework for developers to integrate robust DDoS protection directly into their environments.

Recommendation

Based on the findings, it is recommended that future development focus on enhancing the system's adaptability to emerging and more sophisticated attack vectors by incorporating machine learning-based anomaly detection. Additionally, expanding cloud integration and

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optimizing resource management will improve scalability and performance under high

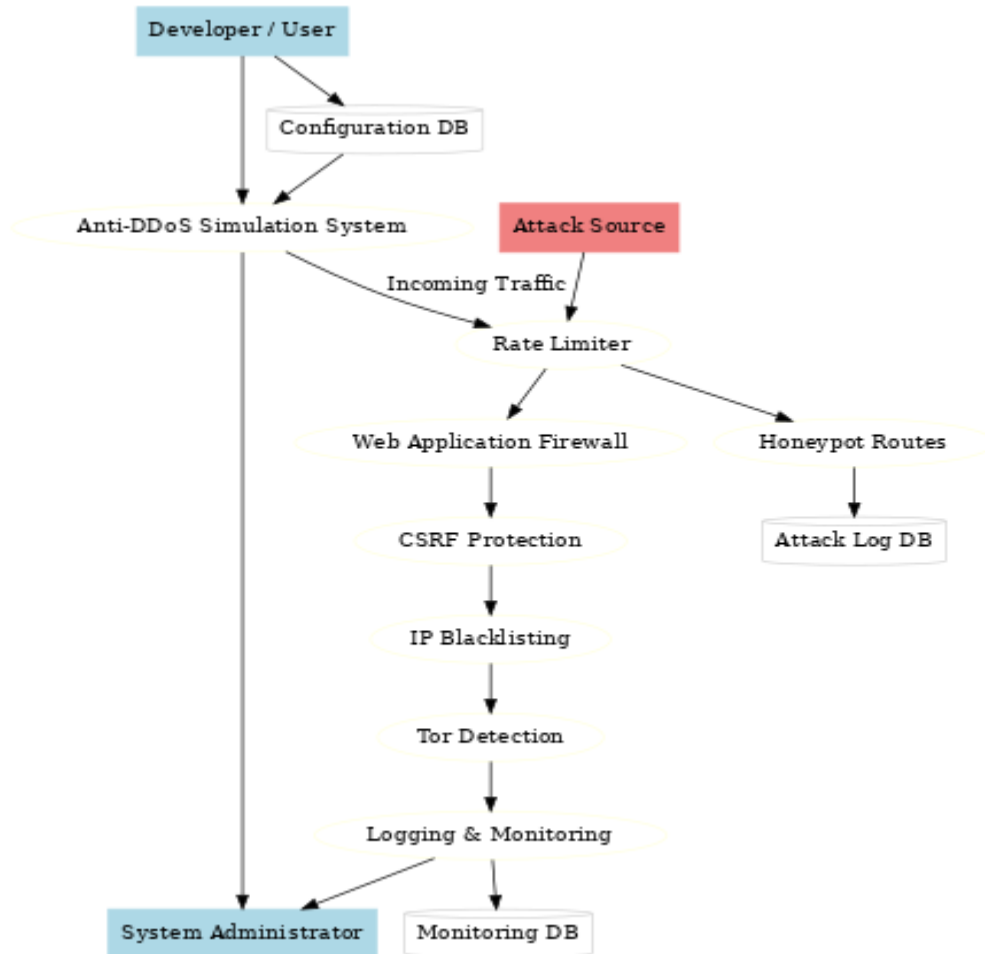
SSSSWSDtraffic loads. It is also advisable to develop user-friendly interfaces and configuration tools to facilitate easier deployment and tuning by non-expert users. Finally, ongoing training and awareness programs should be implemented to keep users informed about evolving threats and best practices in DDoS mitigation.

Final Thoughts

This study underscores the critical need for proactive and comprehensive DDoS protection in today's increasingly connected digital landscape. By combining theoretical knowledge with hands-on simulation, the project not only advanced technical defenses but also empowered users with practical skills to recognize and respond to attacks effectively. As cyber threats continue to evolve, continuous innovation and collaboration between researchers, developers, and users will be essential to maintaining resilient and secure web applications. The Anti-DDoS Protection System developed here represents a meaningful step toward that goal, offering both a valuable educational tool and a foundation for future advancements in cybersecurity.

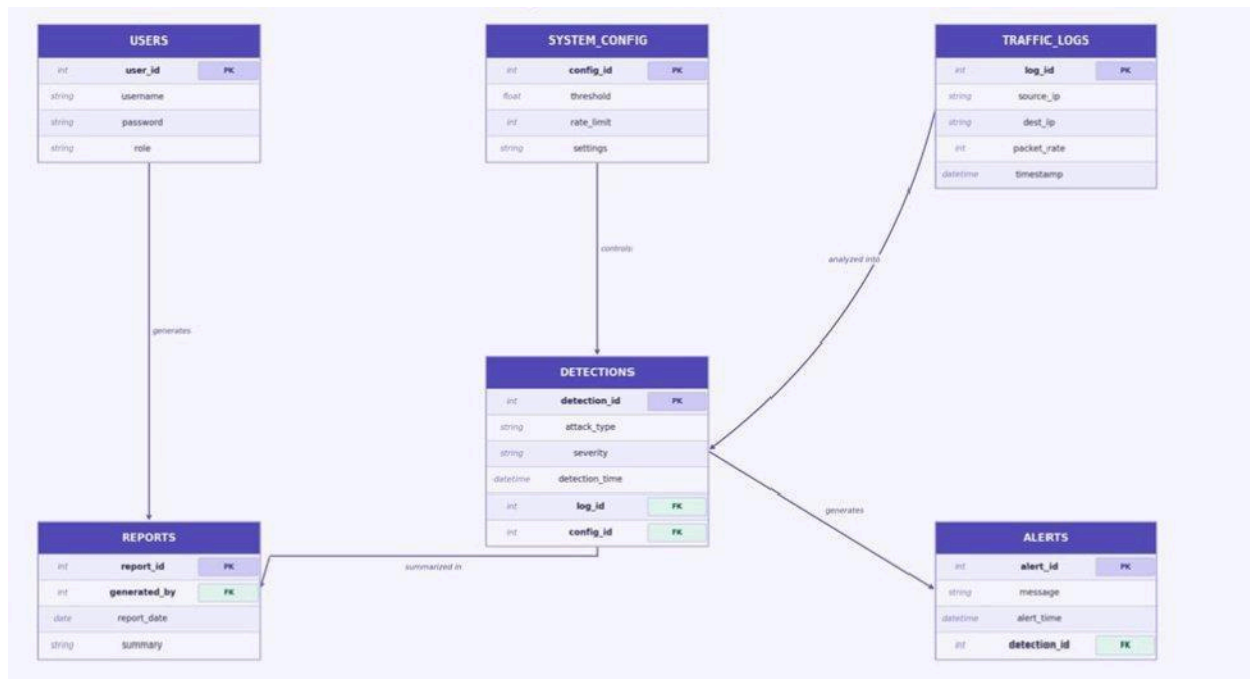
Appendices

Appendix A: Data Flow Diagram



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Appendix B: Entity Relationship Diagram



Code impression management

```
from flask import Flask
from flask_cors import CORS
from config import Config
from database import db
from routes.auth import auth_bp
from routes.profile import profile_bp
from routes.feedback import feedback_bp

def create_app():
```

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```
app = Flask(__name__)
app.config.from_object(Config)

CORS(app, resources={r"/api/*": {"origins": Config.ALLOWED_ORIGINS}})

db.init_app(app)

app.register_blueprint(auth_bp, url_prefix="/api/auth")
app.register_blueprint(profile_bp, url_prefix="/api/profile")
app.register_blueprint(feedback_bp, url_prefix="/api/feedback")

with app.app_context():
    db.create_all() # creates tables if they don't exist

return app

app = create_app()

if __name__ == "__main__":
    app.run(debug=True, port=5000)
```